

Minimal Structural Sufficiency:

From Winning Tickets to Extreme Neuroplasticity

Abstract

A long-standing assumption in both neuroscience and artificial intelligence is that cognitive or inferential capacity scales with the amount of underlying physical substrate. Larger brains or denser networks are implicitly treated as more capable systems. However, recent developments challenge this view from two independent directions: the Lottery Ticket Hypothesis in artificial neural networks and rare but well-documented clinical cases of extreme hydrocephalus with preserved cognitive function.

In this work, we argue that both phenomena point to the same structural principle: functional systems do not require full substrate utilization, but only a minimal structurally sufficient configuration capable of sustaining inference. We formalize this principle as convergence toward a structural minimum (denoted $F_{\min}$), and show how early constraint-driven development enables stable cognition near this boundary. The analysis has implications for localizationist neuroscience, redundancy-based explanations, and theories of consciousness that rely on fixed anatomical correlates.

1. Introduction

The intuition that “more structure implies more function” is deeply embedded in scientific reasoning. In neuroscience, this intuition appears as volume-based correlates of intelligence, modular localization of cognitive faculties, and assumptions about irreducible neural requirements for awareness. In artificial intelligence, it manifests as the scaling paradigm: larger models, more parameters, and denser representations are assumed to yield superior performance.

Yet, both fields have produced anomalies that resist this intuition.

In machine learning, the Lottery Ticket Hypothesis (LTH) demonstrates that dense neural networks contain sparse subnetworks that, when trained in isolation, achieve performance comparable to the full model. In clinical neuroscience, rare cases of severe hydrocephalus show individuals missing the majority of cortical tissue while maintaining near-normal cognitive and social functioning.

These anomalies are typically treated as curiosities, exceptions, or engineering tricks. This paper argues instead that they reveal a deeper structural principle shared across domains.

2. The French Hydrocephalus Case as a Boundary Condition

In 2007, Feuillet et al. reported a striking neurological case: a middle-aged man with massive ventricular enlargement and an extremely thin cortical mantle, yet exhibiting relatively preserved cognitive function and a stable social life. Neuroimaging revealed that approximately 90% of the cranial volume was occupied by cerebrospinal fluid, with only a small fraction of cortical tissue remaining.

Despite this, the patient:

- developed normally,
- maintained employment,
- formed a family,
- and showed only mildly reduced IQ scores.

Crucially, this condition was present from early development. The brain did not lose tissue after maturation; it developed under a severe and persistent structural constraint.

This case poses a fundamental challenge to:

- strict localizationist theories,
- volumetric explanations of cognition,
- and models that assume a direct proportionality between neural mass and function.

Rather than indicating that “any brain region can do anything,” the case suggests something more precise: cognitive function can stabilize far below expected structural capacity, provided development occurs entirely within that constraint.

3. Winning Tickets as Structural, Not Quantitative, Minima

The Lottery Ticket Hypothesis, introduced by Frankle and Carbin, states that within a randomly initialized dense neural network exists a sparse subnetwork—a “winning ticket”—that can be trained to achieve comparable performance to the original network.

Key properties of winning tickets include:

- they are not obvious a priori,
- they are structurally specific, not arbitrary,
- their sufficiency emerges through training dynamics,
- and once identified, the remaining parameters become functionally irrelevant for inference.

Importantly, winning tickets are **not** simply compressed versions of a trained model. They represent a minimal configuration that is *structurally sufficient* to support the learned function.

This reframes model size not as a requirement for inference, but as a search space that enables the discovery of minimal functional structure.

4. Plasticity as Early Convergence to $F_{\min}$

Neuroplasticity is often described in qualitative terms: adaptation, compensation, or reorganization. However, the hydrocephalus case suggests a more formal interpretation.

When a system develops under extreme structural constraint, plasticity operates not as recovery from loss, but as **primary convergence** toward a viable functional minimum.

In the French case:

- there was no pruning of excess structure,
- no late-stage compensation for damage,
- and no transition from a richer to a poorer substrate.

Instead, the cognitive system formed entirely within a severely limited structural envelope. This prevented dependence on unavailable substructures and forced early stabilization near the minimal configuration capable of sustaining learning, inference, and action.

In machine learning terms, this is equivalent to training directly on the winning ticket, without ever instantiating the full dense model.

5. Phase Transitions Near Structural Limits

Both artificial and biological systems exhibit phase-like behavior near critical structural thresholds.

Above a certain capacity, systems are:

- robust,
- redundant,
- and tolerant to perturbations.

Below that threshold, functionality collapses.

The hydrocephalus case and winning ticket networks both operate near this boundary. What distinguishes them from failure cases is not the amount of structure retained, but the manner in which development or training occurs relative to the boundary.

In this sense, cognition is not destroyed by proximity to minimal structure; it is destabilized only when systems are *forced below* their structural minimum after dependency on richer structure has formed.

6. Implications for Localizationist Neuroscience

Traditional localizationist models assume that specific cognitive functions require specific anatomical regions. The case discussed here does not deny regional specialization, but it undermines claims of strict necessity.

If cognitive function can be sustained by radically reduced and reorganized neural substrates, then:

- anatomical location is contingent,
- functional relations matter more than physical placement,
- and explanations based solely on region-function mappings are incomplete.

This does not imply that “the brain does not matter,” but that what matters is **structural sufficiency**, not anatomical completeness.

6.1. Implications for Theories of Consciousness

The hydrocephalus case is particularly unsettling for theories of consciousness that depend on fixed anatomical localization or on the existence of a critical minimum of neural mass. The patient described by Feuillet et al. (2007) retained an estimated 10–20% of typical brain volume, yet exhibited stable consciousness, social integration, and functional independence.

This observation challenges any model that asserts that consciousness requires:

- a specific cortical region,
- a fixed set of neural hubs (e.g., prefrontal cortex, thalamus),
- or a minimum absolute number of integrated neurons.

If such requirements were strictly necessary, consciousness should collapse under such extreme structural reduction. It does not.

One possible interpretation is extreme developmental plasticity. Because the condition was present from early infancy, neural organization adapted continuously within a severely constrained structural envelope. Functions that typically recruit large volumes of neural tissue appear to have been compressed into a thin remaining cortical mantle. As Cleeremans notes, this suggests that plasticity is far more pervasive than traditionally assumed.

However, plasticity alone is not a complete explanation. What matters is not merely reorganization, but **early convergence toward a viable structural configuration capable of sustaining learning, inference, and self-modeling**. Consciousness, in this view, does not depend on absolute neural mass, but on whether the system reaches and stabilizes above a minimal structural threshold.

This directly undermines “strong neuronalist” theories that equate consciousness with hardware quantity or fixed anatomical substrates. Instead, it favors accounts that emphasize:

- dynamic connectivity,
- learning-driven organization,
- and distributed functional coherence.

Importantly, the patient does not exhibit consciousness as a fragile or compensatory state. There is no evidence of continuous effort to overcome deficit. Rather, coherence persists naturally, suggesting that the remaining structure functions not as a damaged system, but as a **complete system operating near its minimal sufficient configuration**.

7. Relation to Structural Limit Theories (TNA Perspective)

Within the framework of the Theory of Necessary Assumptions (TNA), the phenomena described here can be interpreted as manifestations of a deeper constraint: systems capable of inference require a minimal structural configuration $F_{\min}$, below which functionality is impossible and above which additional structure becomes non-essential.

The hydrocephalus case and winning ticket networks occupy the same explanatory category:

- both demonstrate convergence to $F_{\min}$,
- both reveal that excess structure primarily facilitates discovery, not operation,
- and both show that inference is governed by structural necessity, not material abundance.

This interpretation is structural, not metaphysical. It does not posit cognition independent of physical substrate, but rather identifies the minimal conditions under which substrate supports inference.

From a TNA perspective, this case illustrates the distinction between observable structure (N_o) and latent structural necessity (N_i). The observable brain (N_o) is drastically reduced, yet the system remains coherent. This implies the presence of an irreducible structural support—learning dynamics, plastic reconfiguration, and functional organization—that cannot be inferred directly from anatomical volume alone. Consciousness, therefore, does not “emerge” from remaining matter as a passive consequence, but is sustained by processes that precede and organize material instantiation.

8. Conclusion

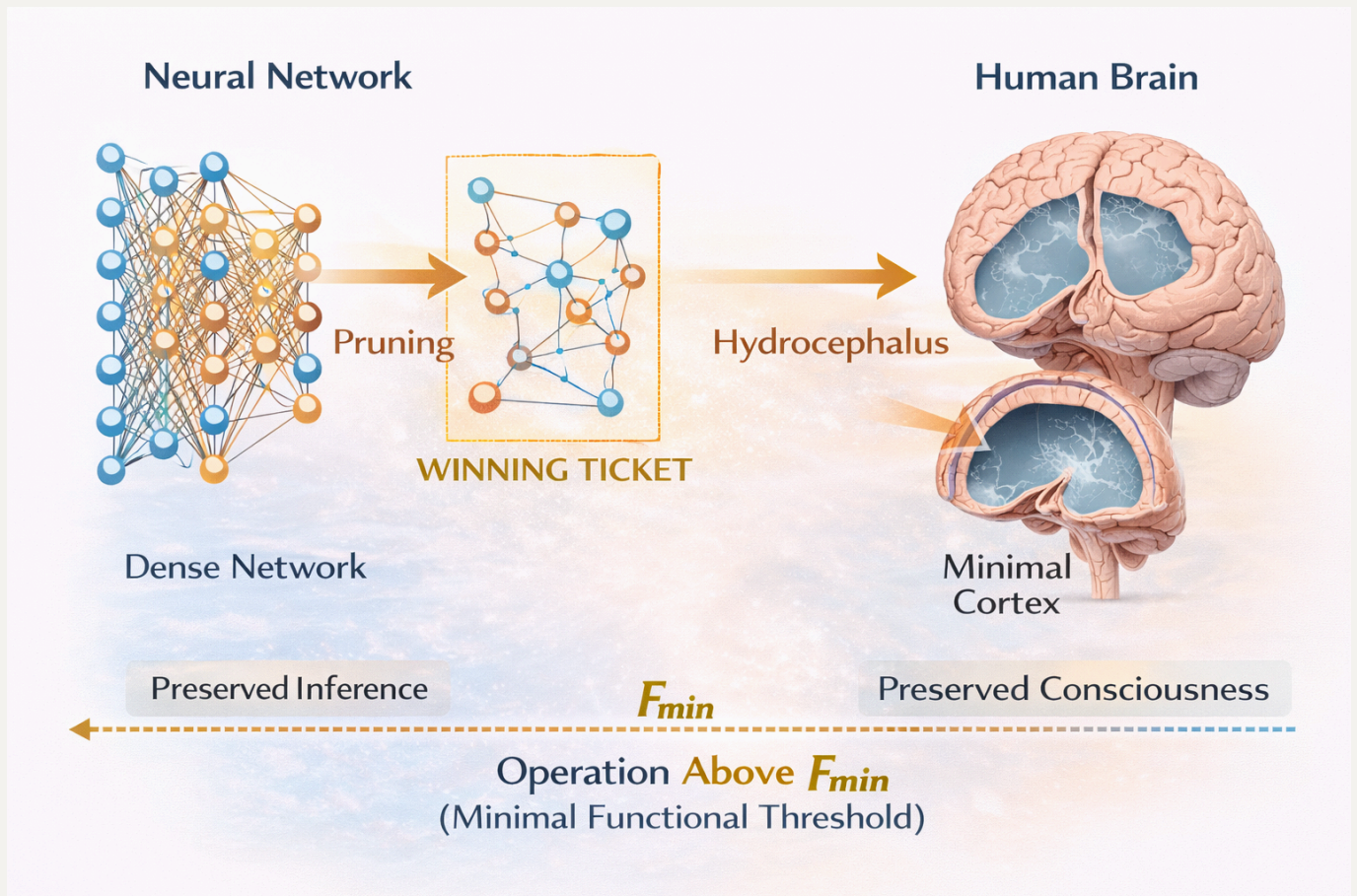
The convergence between extreme neuroplasticity and sparse neural networks suggests that functional intelligence—biological or artificial—is governed by structural minima rather than maximal utilization of resources.

The French hydrocephalus case is not a curiosity to be explained away, but a boundary condition that exposes the limits of current explanatory frameworks. When viewed alongside the Lottery Ticket Hypothesis, it points toward a unifying principle: **function emerges when minimal structural sufficiency is met, not when maximal structure is present.**

Understanding cognition, therefore, requires shifting focus from quantity to structure, from completeness to necessity.

Appendix A

Diagram: Functional Reduction - preserved connections



“The same conceptual diagram is reused here under a different interpretative role, emphasizing its status as a structural invariant rather than a domain-specific illustration.”

Explanation of Appendix A — Conceptual Diagram

(Winning Ticket ↔ Brain Analogy)

This diagram compares two apparently unrelated systems—a neural network and a human brain—to illustrate a shared structural principle: **functional sufficiency above a minimal threshold**.

Left Side: Neural Network (Machine Learning)

1. Dense Network

- Modern neural networks are typically trained with a very large number of parameters.
- Most of these parameters are not strictly necessary for correct inference.

2. Pruning Process

- Through pruning (as formalized in the Lottery Ticket Hypothesis), up to 90% of parameters can be removed.
- What remains is a sparse subnetwork.

3. Winning Ticket

- This minimal subnetwork preserves the original network's inference performance.
- Crucially, functionality depends not on size, but on **structural organization**.
- The system continues to operate **above a minimal functional threshold**, denoted $F_{\min}$.

4. Preserved Inference

- Despite massive reduction, the system still performs its task reliably.
 - The removed parameters were *supportive*, not *essential*.
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Right Side: Human Brain (Extreme Hydrocephalus Case)

1. Typical Brain

- Standard neurological theories often assume that consciousness and cognition require a large mass of neural tissue.

2. Hydrocephalus

- In extreme hydrocephalus, ventricular expansion compresses the brain, leaving only a thin cortical layer.
- In the documented French case, approximately 80–90% of brain volume was absent.

3. Minimal Cortex

- Despite this drastic reduction, the individual:
 - Lived independently
 - Worked as a civil servant
 - Maintained social relationships

- Exhibited stable consciousness and functional cognition

4. Preserved Consciousness

- Conscious experience persists even when neural mass is drastically reduced.
 - This challenges theories that tie consciousness to a fixed location or critical neuronal quantity.
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Shared Structural Axis: Operation Above $F_{\min}$

The dashed horizontal line labeled “**Operation Above $F_{\min}$** ” represents a shared principle:

- Both systems remain functional **as long as they stay above a minimal structural threshold.**
 - This threshold is not defined by size or quantity, but by **organization, connectivity, and learned structure.**
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Structural Interpretation (TNA Framework)

Within the **Theory of Axiomatic Necessity (TNA)**:

- The observable system ($\mathcal{N}_0$) may be drastically reduced.
- Functional coherence persists due to a **latent structural support** ($\mathcal{N}_1$), enabled by:
 - Plasticity
 - Learning
 - Reorganization
 - Boundary-driven adaptation

The diagram visually supports the claim that:

Coherence does not scale with material volume, but with structural sufficiency relative to $F_{\min}$.

Why This Matters

- **For AI:**
Efficiency is not about more parameters, but about identifying structurally necessary ones.
- **For Neuroscience and Consciousness:**
Consciousness cannot be reduced to neural mass or specific locations alone.
- **For Theory (TNA):**
Both systems exemplify the same principle:

Minimal but well-organized structure is sufficient for sustained coherence.

One-Sentence Summary

The diagram shows that both artificial and biological intelligence can lose most of their material substrate and still function, as long as they remain above a minimal structural threshold $F_{\min}$ —supporting a structural, not material, account of cognition and consciousness.

References

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